

Notes on The Riemann Integral

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March 2026

I Partitions and Tagged Partitions

Definition. If $I = [a, b]$ is a closed bounded interval, then a **partition** of I is a finite, ordered set $\mathcal{P} := (x_0, x_1, \dots, x_{n-1}, x_n)$ of points such that

$$a = x_0 < x_1 < \dots < x_{n-1} < x_n = b$$

partitions are used to divide I into non overlapping subintervals $I_1 = [x_0, x_1], I_2 = [x_1, x_2], \dots$. We can denote a partition by

$$\mathcal{P} = \left\{ [x_{i-1}, x_i] \right\}_{i=1}^n$$

We can define a **norm** of a partition to be

$$\|\mathcal{P}\| := \max\{x_1 - x_0, \dots, x_n - x_{n-1}\}$$

The norm of $\mathcal{P}$ is the length of the largest subinterval. If a point t_i is selected from each subinterval $I_i = [x_{i-1}, x_i]$ for $i = 1, 2, 3, \dots, n$ then the points are called **tags** of the subintervals. A set of ordered pairs of subintervals and their corresponding tags is called a tagged partition

$$\dot{\mathcal{P}} := \left\{ [x_{i-1}, x_i], t_i \right\}_{i=1}^n$$

We can choose a tag in any way we want to. Given a tagged partition we can define the **Riemann Sum** of a function corresponding to $\dot{\mathcal{P}}$ as

$$S(f; \dot{\mathcal{P}}) := \sum_{i=1}^n f(t_i)(x_i - x_{i-1}) \tag{1}$$

Note that we use $\dot{\mathcal{P}}$ also when we are talking about a subset of a partition. If f positive, then the Riemann Sum is the sum of the areas of rectangles whose bases are $[x_{i-1}, x_i]$ and heights determined by $f(t_i)$.

[†]Definitions and example problems from MATH 42002: Analysis II instructed by Dr. Zachary Chase and *Introduction to Real Analysis* by Robert G. Bartle and Donald R. Sherbert

II The Riemann Integral

A function $f : [a, b] \rightarrow \mathbb{R}$ is said to be Riemann integrable on $[a, b]$ if there exists a number $L \in \mathbb{R}$ such that for every $\varepsilon > 0$, there exists $\delta_\varepsilon > 0$ such that if $\dot{\mathcal{P}}$ is a tagged partition of $[a, b]$ with $||\dot{\mathcal{P}}|| < \delta_\varepsilon$ then

$$|S(f; \dot{\mathcal{P}}) - L| < \varepsilon$$

We denote that a function f is Riemann integrable by $f \in \mathcal{R}[a, b]$ where $\mathcal{R}[a, b]$ is the set of all Riemann integrable functions on $[a, b]$. We also can use the traditional integral symbol as well.

Theorem. If $f \in \mathcal{R}[a, b]$ then the value of the integral is uniquely determined.

Examples

a) Every constant function on $[a, b]$ is Riemann Integrable.

Proof. If we let $f(x) = k$ then the Riemann sum becomes

$$S(f; \dot{\mathcal{P}}) := \sum_{i=1}^n k(x_i - x_{i-1}) = k(b - a)$$

So then $\forall \varepsilon > 0$, we can choose $\delta_\varepsilon = 1$ such that if $||\dot{\mathcal{P}}|| < \delta_\varepsilon$ then

$$|S(f; \dot{\mathcal{P}}) - k(b - a)| = 0 < \varepsilon$$

Therefore, $f \in \mathcal{R}[a, b]$ $\square$

b) Let $h(x) = x$ for $x \in [0, 1]$. Show that h is Riemann Integrable.

Proof. One trick we can use is to choose a tag that is the midpoint of the interval we partition. So, if $\mathcal{Q}$ is a partition of $I = [x_{i-1}, x_i]$ then we choose tags to be the midpoint $q_i = \frac{1}{2}(x_{i-1} + x_i)$ then we obtain tagged partition $\dot{\mathcal{Q}} = \left\{ [x_{i-1}, x_i], q_i \right\}_{i=1}^n$. Then, the Riemann Sum is

$$S(h; \dot{\mathcal{Q}}) = \sum_{i=1}^n \frac{1}{2}(x_i + x_{i-1})(x_i - x_{i-1}) = \frac{1}{2}(x_i^2 - x_{i-1}^2) = \frac{1}{2}$$

We obtain $1/2$ by substituting in for $0, 1$. Taking $\varepsilon > 0$ and $\delta_\varepsilon = \varepsilon$, Let $\dot{\mathcal{P}}$ to be a tagged partition of $[0, 1]$ with $||\dot{\mathcal{P}}|| < \delta_\varepsilon$. Also let $\dot{\mathcal{Q}}$ have the same partition points. Then $|t_i - q_i| < \delta_\varepsilon$ since both $t_i, q_i \in I$. By the triangle inequality,

$$\begin{aligned} |S(h; \dot{\mathcal{P}}) - S(h; \dot{\mathcal{Q}})| &= \left| \sum_{i=1}^n t_i(x_i - x_{i-1}) - \sum_{i=1}^n q_i(x_i - x_{i-1}) \right| \leq \sum_{i=1}^n |t_i - q_i|(x_i - x_{i-1}) \\ &< \delta_\varepsilon \sum_{i=1}^n (x_i - x_{i-1}) = \delta_\varepsilon (x_n - x_0) = \delta_\varepsilon \end{aligned}$$

and since $S(h; \dot{\mathcal{Q}}) = 1/2$ then

$$\left| S(h; \dot{\mathcal{P}}) - \frac{1}{2} \right| < \delta_\varepsilon = \varepsilon$$

Therefore, $f \in \mathcal{R}[0, 1]$ $\square$

Properties of the Integral

If f, g are Riemann integrable, then if $k \in \mathbb{R}$, the function kf is also Riemann Integrable and

$$\int_a^b kf = k \int_a^b f$$

Proof. $S(kf; \dot{\mathcal{P}}) = kS(f; \dot{\mathcal{P}})$ $\square$.

The function $f + g \in \mathcal{R}[a, b]$ and

$$\int_a^b f + g = \int_a^b f + \int_a^b g$$

Proof. $S(f + g; \dot{\mathcal{P}}) = S(f; \dot{\mathcal{P}}) + S(g; \dot{\mathcal{P}})$. Let $\varepsilon > 0$, then pick $\delta_\varepsilon = \min\{\delta_f, \delta_g\}$. Since f, g Riemann integrable, then there exists $\delta_f, \delta_g > 0$ such that when $\|\dot{\mathcal{P}}\| < \delta_f$ then $|S(f; \dot{\mathcal{P}}) - \int_a^b f| < \varepsilon$ and when $\|\dot{\mathcal{P}}\| < \delta_g$ then $|S(g; \dot{\mathcal{P}}) - \int_a^b g| < \varepsilon$. Then by the Triangle Inequality we have,

$$\left| S(f + g, \dot{\mathcal{P}}) - \left(\int_a^b f + \int_a^b g \right) \right| \leq \left| S(f + g, \dot{\mathcal{P}}) - \int_a^b f \right| + \left| S(f + g, \dot{\mathcal{P}}) - \int_a^b g \right| < 2\varepsilon$$

$\square$

If $f \leq g \forall x \in [a, b]$ then

$$\int_a^b f \leq \int_a^b g$$

Proof.

$$\int_a^b f - \varepsilon < S(f; \dot{\mathcal{P}}) \leq S(g; \dot{\mathcal{P}}) < \int_a^b g + \varepsilon \implies \int_a^b f < \int_a^b g + \varepsilon \implies \int_a^b f < \int_a^b g$$

$\square$

The Boundedness Theorem. If $f \in \mathcal{R}[a, b]$ then f is bounded on $[a, b]$.

Examples.

a) If $f \in \mathcal{R}[a, b]$ and $|f| \leq M$, then $|\int_a^b f| \leq M(b - a)$

Proof. $\forall \varepsilon > 0, \exists \delta_\varepsilon < 0$ such that if $\|\dot{\mathcal{P}}\| < \delta_\varepsilon$ then

$$\begin{aligned} \left| \sum_{i=1}^n f(t_i)(x_i - x_{i-1}) - L \right| < \varepsilon &\implies |L| < \left| \sum_{i=1}^n f(t_i)(x_i - x_{i-1}) \right| + \varepsilon \leq \sum_{i=1}^n |f(t_i)|(x_i - x_{i-1}) + \varepsilon \\ &\leq \sum_{i=1}^n M(x_i - x_{i-1}) + \varepsilon = M(b - a) + \varepsilon \end{aligned}$$

$\square$

b) Let $g : [0, 1] \rightarrow \mathbb{R}$ Also let $g(x) = 0$ if x rational and $1/x$ if x irrational. Prove g is not Riemann Integrable.

Proof. g is not bounded (since if x irrational, the limit as x goes to zero from the right goes to infinity) and is therefore not Riemann integrable on $[0, 1]$ by the contrapositive of The Boundedness Theorem. $\square$

We conclude this section with a note that Thomae's function is Riemann Integrable and that the Dirichlet function is not Riemann integrable on $[0, 1]$.

III Riemann Integrable Functions

The Cauchy Criterion. A function $f : [a, b] \rightarrow \mathbb{R}$ is in $\mathcal{R}[a, b]$ iff for every $\varepsilon > 0$ there exists $\eta_\varepsilon > 0$ such that if $\dot{\mathcal{P}}$ and $\dot{\mathcal{Q}}$ are any tagged partitions of $[a, b]$ with $\|\dot{\mathcal{P}}\| < \eta_\varepsilon$ and $\|\dot{\mathcal{Q}}\| < \eta_\varepsilon$, then

$$|S(f; \dot{\mathcal{P}}) - S(f; \dot{\mathcal{Q}})| < \varepsilon$$

In essence, we are saying that the Riemann sums (and partitions) are close to each other.

The Squeeze Theorem. Let $f : [a, b] \rightarrow \mathbb{R}$. Then, $f \in \mathcal{R}[a, b]$ iff for every $\varepsilon > 0$ there exist functions α_ε and ω_ε in $\mathcal{R}[a, b]$ with

$$\alpha_\varepsilon(x) \leq f(x) \leq \omega_\varepsilon(x), \quad \forall x \in [a, b]$$

and

$$\int_a^b (\omega_\varepsilon - \alpha_\varepsilon) < \varepsilon$$

Proof. ($\implies$): For all $\varepsilon > 0$, we can simply take $\alpha_\varepsilon = \omega_\varepsilon = f$.

For the opposite direction, let $\varepsilon > 0$. Since $\alpha_\varepsilon, \omega_\varepsilon$ are Riemann integrable, then there exists a $\delta_\varepsilon > 0$ such that if $\dot{\mathcal{P}}$ is any tagged partition, with $\|\dot{\mathcal{P}}\| < \delta_\varepsilon$ then we have

$$|S(\alpha_\varepsilon; \dot{\mathcal{P}}) - \int_a^b \alpha_\varepsilon| < \varepsilon, \quad |S(\omega_\varepsilon; \dot{\mathcal{P}}) - \int_a^b \omega_\varepsilon| < \varepsilon$$

Which we can use to obtain the following inequalities

$$\int_a^b \alpha_\varepsilon - \varepsilon < S(\alpha_\varepsilon; \dot{\mathcal{P}}), \quad S(\alpha_\varepsilon; \dot{\mathcal{P}}) < \int_a^b \omega_\varepsilon + \varepsilon$$

Since $\alpha_\varepsilon(x) \leq f(x) \leq \omega_\varepsilon(x)$ and all three functions are Riemann integrable we have $S(\alpha_\varepsilon; \dot{\mathcal{P}}) < S(f; \dot{\mathcal{P}}) < S(\omega_\varepsilon; \dot{\mathcal{P}})$ from where

$$\int_a^b \alpha_\varepsilon - \varepsilon < S(f; \dot{\mathcal{P}}) < \int_a^b \omega_\varepsilon + \varepsilon$$

and if $\dot{\mathcal{Q}}$ is any tagged partition with $\|\dot{\mathcal{Q}}\| < \delta_\varepsilon$ then

$$\int_a^b \alpha_\varepsilon - \varepsilon < S(f; \dot{\mathcal{Q}}) < \int_a^b \omega_\varepsilon + \varepsilon$$

Subtracting the inequalities and using that $\int_a^b (\omega_\varepsilon - \alpha_\varepsilon) < \varepsilon$ then

$$|S(\alpha_\varepsilon; \dot{\mathcal{P}}) - S(f; \dot{\mathcal{Q}})| < \int_a^b \omega_\varepsilon - \int_a^b \alpha_\varepsilon + 2\varepsilon = \int_a^b (\omega_\varepsilon - \alpha_\varepsilon) + 2\varepsilon < 3\varepsilon$$

Therefore, by the Cauchy Criterion, $f \in \mathcal{R}[a, b]$. $\square$

Theorem. If $\varphi : [a, b] \rightarrow \mathbb{R}$ is a step function (if it has only a finite number of distinct values, where each value is assumed on one or more subintervals) then $\varphi \in \mathcal{R}[a, b]$

Theorem. If $f : [a, b] \rightarrow \mathbb{R}$ is continuous on $[a, b]$ then f is Riemann integrable on $[a, b]$.

Theorem. If $f : [a, b] \rightarrow \mathbb{R}$ is monotone on $[a, b]$ then f is Riemann integrable on $[a, b]$.

The Additivity Theorem. Let $f : [a, b] \rightarrow \mathbb{R}$ and let $c \in (a, b)$. Then $f \in \mathcal{R}[a, b]$ iff its restrictions to $[a, c]$ and $[c, b]$ are both Riemann integrable, i.e.,

$$\int_a^b f = \int_a^c f + \int_c^b f$$

Corollary. If f Riemann integrable and if $[c, d] \subseteq [a, b]$ then the restriction of f to $[c, d]$ is in $\mathcal{R}[c, d]$.

Corollary. If f Riemann integrable and if $a = c_0 < c_1 < \dots < c_m = b$ then the restrictions of f to each subinterval $[c_{i-1}, c_i]$ are Riemann integrable and

$$\int_a^b f = \sum_{i=1}^m \int_{c_{i-1}}^{c_i} f$$

Definition. If $f \in \mathcal{R}[a, b]$ and if $\alpha, \beta \in [a, b]$ with $\alpha < \beta$ we say

$$\int_{\beta}^{\alpha} f := - \int_{\alpha}^{\beta} f \quad \text{and} \quad \int_{\alpha}^{\alpha} f := 0$$

Theorem. If $f \in \mathcal{R}[a, b]$ and if α, β, γ are any numbers in $[a, b]$, then

$$\int_{\alpha}^{\beta} f = \int_{\alpha}^{\gamma} f + \int_{\gamma}^{\beta} f$$

IV The Fundamental Theorem

The Fundamental Theorem of Calculus (First Form). Suppose that there is a finite set E in $[a, b]$ and functions $f, F : [a, b] \rightarrow \mathbb{R}$ such that

1. F is continuous on $[a, b]$
2. $F'(x) = f(x) \quad \forall x \in [a, b] \setminus E$
3. $f \in \mathcal{R}[a, b]$

Then, we have that

$$\int_a^b f = F(b) - F(a)$$

Note that if F is differentiable at every point in $[a, b]$ 1. is automatically satisfied. If it is not defined for some $c \in E$, then we take $f(c) = 0$.

Examples.

Definition. If f is Riemann Integrable, then the function defined by

$$F(z) := \int_a^z f \quad \text{for } z \in [a, b]$$

is called the indefinite integral of f with base point a .

Theorem. The indefinite integral F defined above is continuous on $[a, b]$. In fact, it is also Lipschitz.

The Fundamental Theorem of Calculus (Second Form). Let $f \in \mathcal{R}[a, b]$ and let f be continuous at $c \in [a, b]$. Then the indefinite integral is differentiable at c and $F'(c) = f(c)$. Note that if we let f be continuous on all of $[a, b]$ then we obtain

Theorem. If f continuous on $[a, b]$, then the indefinite integral F is differentiable on $[a, b]$ and $F'(x) = f(x) \forall x \in [a, b]$.

Note that in the above theorem we are saying that if f continuous on $[a, b]$, then its indefinite integral is an anti-derivative of f . Also note that, in general, the indefinite integral need not be an anti-derivative either because the derivative of the indefinite integral does not exist or is not equal to $f(x)$. Consider $f(x) = \text{sgn}$ on $[-1, 1]$. Then, $f \in \mathcal{R}[-1, 1]$ and has indefinite integral $F(x) = |x| - 1$ with base point -1 . But, since $F'(0)$ does not exist, then F is not an anti-derivative of f on $[-1, 1]$. Also consider h to be Thomae's Function. On $[0, 1]$ the indefinite integral H is 0 but when $x \in \mathbb{Q} \cap [0, 1]$ then $H'(x) \neq h(x)$ so H is not an anti-derivative of h on $[0, 1]$.

The Substitution Theorem. Let $J = [\alpha, \beta]$ and let $\varphi : J \rightarrow \mathbb{R}$ be a continuous derivative on J . If $f : I \rightarrow \mathbb{R}$ is a continuous on an interval I containing $\varphi(J)$ then

$$\int_{\alpha}^{\beta} f(\varphi(t)) \cdot \varphi'(t) dt = \int_{\varphi(\alpha)}^{\varphi(\beta)} f(x) dx$$

Note that this is a precursor to the more powerful version of U-Substitution for the generalized Riemann integral.